Etherscan Address Tag Task — Setup & Findings

Goal: Use an address list to query Etherscan for an Ethereum address's name tag and labels, then save the results for analysis.

1. Project Setup

A Windows Python virtual environment was created successfully in `.venv/Scripts/`. Because the VS Code terminal is Git Bash (MINGW64), the correct activation command is:

```
source .venv/Scripts/activate
```

Successful activation is shown by `(.venv)` at the beginning of the terminal prompt.

2. Initial API Attempt

The original V1 request used `https://api.etherscan.io/api`. Etherscan returned a message that the V1 endpoint is deprecated and instructed users to migrate to API V2.

Result: The problem was not Python or the virtual environment; the endpoint was outdated.

3. API V2 Update

The request was changed to the V2 endpoint and Ethereum Mainnet chain ID:

```
https://api.etherscan.io/v2/api
chainid = 1
module = nametag
action = getaddresstag
```

The V2 request successfully reached Etherscan, confirming that the Python request structure and API key were being sent to the current API.

4. Current Result / Blocker

Etherscan returned: `You are trying to access an API Exclusive endpoint. Contact us to upgrade your API plan.`

Interpretation: The current API V2 endpoint `getaddresstag` is restricted for the API plan being used. This is an Etherscan plan/access restriction, not a coding error.

5. Address List Requirement

The task requires an address list because `getaddresstag` operates on one address per call. If the assignment provides a CSV, use that CSV. If no list is provided, a small test list can be created, but real addresses should come from a legitimate data source rather than being invented.

6. Intended Data Pipeline

Address list → validate/deduplicate → call Etherscan for each address → extract name tag + labels → save results to CSV/JSON/database.

7. Recommended Next Step

Do **not** purchase or upgrade an Etherscan plan yet. The task description says the endpoint is free, while the current V2 API reports that it is API-exclusive. The next step is to inspect the complete assignment instructions and determine whether they refer to an older API, require a particular Etherscan plan, or expect another source/method.